

Product Requirements Document: ISP Order Management System

1. Introduction

This document outlines the product requirements for a new Order Management System (OMS) for an Internet Service Provider (ISP). The system will manage the entire lifecycle of customer orders, from internal creation to fulfillment through various Fiber Network Operators (FNOs). It will also handle order status tracking, escalations, communication, and system administration.

2. Project Overview

The current order management process is fragmented and relies on manual interventions, leading to inefficiencies, errors, and a suboptimal customer experience. This project aims to develop a centralized and automated OMS that will streamline order processing, improve operational efficiency, and enhance customer satisfaction. The system will serve as a single source of truth for all order-related information and will integrate seamlessly with both internal systems and external FNO portals.

3. Goals and Objectives

- **Goal:** To create a robust and scalable Order Management System that automates and streamlines the entire order lifecycle for an ISP.
- **Objective 1:** Reduce order processing time by 50%.
- **Objective 2:** Decrease order fallout rate by 75%.
- **Objective 3:** Improve customer satisfaction by providing real-time order status and proactive communication.
- **Objective 4:** Centralize all order-related data and provide a single source of truth.
- **Objective 5:** Enable seamless integration with multiple FNOs.

4. Functional Requirements

4.1. Order Creation and Management

- **Internal Order Creation:** The system shall allow authorized internal users (e.g., sales, support) to create new customer orders. This includes capturing customer details,

service information, and any special instructions.

- **Order Validation:** The system shall automatically validate all new orders to ensure data completeness and accuracy. This includes address validation, service availability checks, and credit checks.
- **Order Enrichment:** The system shall enrich orders with additional information required for fulfillment, such as network-specific parameters and FNO-specific details.
- **Order Tracking:** The system shall provide real-time tracking of all orders throughout their lifecycle. Users shall be able to view the current status of any order, as well as a complete history of all status changes.
- **Order Modification and Cancellation:** The system shall allow authorized users to modify or cancel orders, subject to business rules and FNO policies.

4.2. FNO Integration

- **Multi-FNO Support:** The system shall support integration with multiple FNOs. This includes the ability to send orders to different FNOs based on service location and availability.
- **Dual Application Methods:** The system shall support both automated API-based and manual FNO application processes based on FNO capabilities and business requirements.
- **API-Based FNO Applications:** The system shall automatically submit orders to FNOs that support API integration, including real-time status synchronization and automated processing.
- **Manual FNO Applications:** The system shall route orders requiring manual application to designated Application Administrators through a dedicated inbox system.
- **Application Administrator Inbox:** The system shall provide a dedicated inbox for Application Administrators showing all pending manual FNO applications with priority indicators and aging information.
- **Manual Application Workflow:** The system shall allow Application Administrators to manually apply orders on FNO portals and capture the FNO application/order reference number for tracking purposes.
- **FNO Reference Tracking:** The system shall store and track FNO-provided application and order reference numbers for all applications, enabling end-to-end order tracking.
- **Status Synchronization:** The system shall automatically synchronize order statuses with FNO systems for API-integrated FNOs and allow manual status updates for manually processed applications.

- **FNO Communication:** The system shall facilitate communication with FNOs, including sending and receiving messages, and logging all communication for future reference.

4.3. Workflow and State Management

- **Configurable Workflows:** The system shall provide configurable workflows for different order types (e.g., new install, disconnect, service change).
- **State Machine:** The system shall implement a robust state machine to manage the lifecycle of each order. The state machine shall define all possible order states and the valid transitions between them.
- **Automated State Transitions:** The system shall automate state transitions based on triggers from internal users, FNOs, and other systems.

4.5. Customer Onboarding System

- **New Customer Onboarding:** The system shall provide a comprehensive onboarding workflow for new customers to ensure smooth service activation and customer satisfaction.
- **Trial Customer Management:** The system shall manage customers on 30-day free trials, tracking trial periods and implementing conversion strategies to ensure purchase before trial expiration.
- **Onboarding Workflow Automation:** The system shall automate onboarding tasks including welcome communications, service setup guidance, and progress tracking.
- **Trial Conversion Tracking:** The system shall monitor trial customer engagement and automatically trigger conversion campaigns based on usage patterns and trial timeline.
- **Onboarding Progress Monitoring:** The system shall track onboarding completion rates and identify customers who may need additional support or intervention.
- **Personalized Onboarding Paths:** The system shall support different onboarding workflows based on customer type, service package, and business requirements.

4.6. Escalation and Communication

- **Time-Based Escalation:** The system shall automatically escalate workflow tasks and orders that exceed predefined time thresholds. Escalation rules shall be configurable based on task type, order type, FNO, and other parameters.
- **Hierarchical Escalation:** The system shall support multi-level escalation workflows, escalating first to reporting managers and then to process owners when tasks remain unattended beyond specified timeframes.

- **Manual Escalations:** The system shall allow users to manually escalate orders and tasks when necessary, with proper justification and audit trail.
- **Escalation Notifications:** The system shall send automated notifications to escalation recipients including task details, aging information, and business impact assessment.
- **SLA Monitoring:** The system shall monitor Service Level Agreements (SLAs) for all workflow tasks and trigger escalations before SLA breaches occur.
- **Proactive Communication:** The system shall send proactive communications to customers and internal stakeholders at key milestones in the order lifecycle (e.g., order confirmation, installation scheduled, service activated).
- **Communication Templates:** The system shall provide configurable templates for all communications, including escalation notifications and customer updates.

4.7. System Settings and Policies

- **User Management:** The system shall provide a user management module for creating and managing user accounts. This includes assigning roles and permissions to users.
- **Role-Based Access Control (RBAC):** The system shall implement RBAC to ensure that users only have access to the information and functionality that is relevant to their role.
- **Policy Management:** The system shall provide a policy management module for defining and managing business rules and policies. This includes policies for order validation, escalation, and communication.
- **Audit Trail:** The system shall maintain a complete audit trail of all changes made to orders, users, and policies.

5. Non-Functional Requirements

5.1. Performance

- The system shall be able to process up to 1,000 orders per day.
- The system shall have a response time of less than 2 seconds for all user interactions.
- The system shall be able to handle large volumes of data without degradation in performance.

5.2. Scalability

- The system shall be designed to scale horizontally to accommodate future growth in order volume and user load.

- The system shall be able to support the addition of new FNOs and services without requiring significant architectural changes.

5.3. Reliability

- The system shall have an uptime of 99.9%.
- The system shall be resilient to failures and shall be able to recover quickly from any outages.
- The system shall have robust error handling and logging mechanisms.

5.4. Security

- The system shall be secure and shall protect all customer and order data from unauthorized access.
- The system shall comply with all relevant security standards and regulations.
- The system shall encrypt all sensitive data, both in transit and at rest.

5.5. Usability

- The system shall be easy to use and shall have an intuitive user interface.
- The system shall provide clear and concise error messages and help documentation.
- The system shall be accessible to users with disabilities.

6. System Architecture

6.1. High-Level Architecture

The ISP Order Management System will be built using a microservices architecture to ensure scalability, maintainability, and flexibility. The system will consist of the following key components:

- **API Gateway:** Serves as the single entry point for all client requests and handles authentication, authorization, and routing.
- **Order Service:** Manages the core order lifecycle, including creation, validation, modification, and tracking.
- **Customer Onboarding Service:** Manages new customer onboarding workflows and trial customer conversion processes.
- **FNO Integration Service:** Handles communication with external FNO systems through both APIs and manual application workflows.

- **Application Administrator Service:** Manages the inbox system for manual FNO applications and tracks application administrator tasks.
- **Workflow Engine:** Manages order workflows and state transitions using a configurable business process management (BPM) engine.
- **Escalation Service:** Handles time-based escalations and hierarchical escalation workflows to reporting managers and process owners.
- **Notification Service:** Handles all internal and external communications, including customer notifications and system alerts.
- **User Management Service:** Manages user accounts, roles, and permissions.
- **Audit Service:** Maintains comprehensive audit logs of all system activities.
- **Reporting Service:** Generates reports and analytics for business intelligence.

6.2. Integration Points

The system will integrate with the following external systems:

- **FNO Systems:** Multiple fiber network operator platforms for order submission and status updates.
- **CRM System:** Customer relationship management system for customer data and interaction history.
- **Billing System:** Financial system for billing and invoicing processes.
- **Inventory Management System:** Equipment and resource tracking system.
- **Notification Platforms:** Email and SMS services for customer communications.
- **Third-party APIs:** External services such as address validation and credit checking.

6.3. Data Architecture

The system will use a combination of relational and NoSQL databases:

- **PostgreSQL:** Primary database for transactional data including orders, customers, and user information.
- **MongoDB:** Document database for storing flexible data structures such as FNO-specific configurations and communication logs.
- **Redis:** In-memory cache for session management and frequently accessed data.

7. User Stories and Acceptance Criteria

7.1. Order Creation

User Story: As a sales representative, I want to create a new customer order so that I can initiate the service provisioning process.

Acceptance Criteria:

- I can enter customer details including name, contact information, and service address.
- I can select the desired service type and package.
- The system validates the entered information and highlights any errors.
- I can save the order as a draft or submit it for processing.
- The system assigns a unique order number upon creation.

7.2. Order Tracking

User Story: As a customer support agent, I want to track the status of customer orders so that I can provide accurate updates to customers.

Acceptance Criteria:

- I can search for orders using various criteria (order number, customer name, phone number).
- I can view the current status and complete history of any order.
- I can see estimated completion dates and any delays or issues.
- I can add notes and comments to orders for future reference.

7.3. FNO Integration

User Story: As an operations manager, I want orders to be automatically submitted to the appropriate FNO so that the provisioning process can begin without manual intervention.

Acceptance Criteria:

- The system automatically determines the correct FNO based on service location.
- Orders are submitted to the FNO's system using their preferred method (API or portal).
- The system receives and processes status updates from the FNO.
- Any errors or rejections from the FNO are logged and escalated appropriately.

7.4. Escalation Management

User Story: As an operations manager, I want orders that are delayed or have issues to be automatically escalated so that they receive appropriate attention.

Acceptance Criteria:

- The system automatically escalates orders that exceed predefined time thresholds.

- Escalation rules can be configured based on order type, FNO, and customer tier.
- Escalated orders are assigned to appropriate personnel based on their role and availability.
- Escalation notifications are sent to relevant stakeholders.

7.5. Customer Onboarding

User Story: As a customer success manager, I want to manage the onboarding process for new customers so that they have a smooth experience and successful service activation.

Acceptance Criteria:

- I can view all customers currently in the onboarding process.
- I can track onboarding progress and completion status for each customer.
- I can assign onboarding tasks to team members and track their completion.
- I can send personalized welcome communications and setup instructions.
- I can identify customers who may need additional support during onboarding.

7.6. Trial Customer Conversion

User Story: As a sales manager, I want to track trial customers and implement conversion strategies so that they purchase services before their 30-day trial expires.

Acceptance Criteria:

- I can view all customers currently on trial with their remaining trial days.
- I can see trial customer engagement metrics and usage patterns.
- The system automatically sends conversion reminders at predefined intervals (e.g., day 7, 14, 21, 28).
- I can create targeted conversion campaigns for different customer segments.
- I can track conversion rates and identify successful conversion strategies.

7.7. Manual FNO Application Management

User Story: As an Application Administrator, I want to manage manual FNO applications through a dedicated inbox so that I can efficiently process orders that require manual submission to FNO portals.

Acceptance Criteria:

- I can view all pending manual FNO applications in my dedicated inbox.
- I can see priority indicators and aging information for each application.

- I can access order details and customer information for each application.
- I can manually apply orders on FNO portals and capture the FNO reference number.
- I can update order status and add notes about the application process.
- The system tracks my application activities and maintains an audit trail.

7.8. Time-Based Escalation Management

User Story: As a reporting manager, I want to receive escalated tasks when my team members don't complete workflow items within specified timeframes so that I can ensure timely order processing.

Acceptance Criteria:

- I receive automated notifications when tasks assigned to my team exceed time thresholds.
- I can view escalated tasks with aging information and business impact details.
- I can reassign escalated tasks to other team members or take ownership myself.
- I can configure escalation rules and timeframes for different task types.
- The system escalates to process owners if I don't address escalated tasks within my timeframe.

7.9. System Administration

User Story: As a system administrator, I want to manage user accounts and permissions so that I can control access to the system.

Acceptance Criteria:

- I can create, modify, and deactivate user accounts.
- I can assign roles and permissions to users based on their job function.
- I can view audit logs of all user activities.
- I can configure system settings and business rules.

8. Technical Specifications

8.1. Technology Stack

- **Frontend:** React.js with TypeScript for the web application
- **Backend:** Node.js with Express.js framework

- **Database:** PostgreSQL for primary data, MongoDB for document storage, Redis for caching
- **Message Queue:** Apache Kafka for asynchronous communication between services
- **Workflow Engine:** Camunda BPM for business process management
- **API Documentation:** OpenAPI/Swagger for API specification
- **Monitoring:** Prometheus and Grafana for system monitoring and alerting
- **Logging:** ELK Stack (Elasticsearch, Logstash, Kibana) for centralized logging

8.2. Development Standards

- **Code Quality:** ESLint and Prettier for code formatting and quality checks
- **Testing:** Jest for unit testing, Cypress for end-to-end testing
- **Version Control:** Git with GitFlow branching strategy
- **CI/CD:** Jenkins or GitHub Actions for continuous integration and deployment
- **Documentation:** Comprehensive API documentation and user guides

9. Success Metrics

- **Order Processing Time:** Reduce average order processing time by 50%
- **Order Accuracy:** Achieve 99% order accuracy rate
- **Customer Satisfaction:** Maintain customer satisfaction score above 4.5/5
- **System Uptime:** Achieve 99.9% system availability
- **User Adoption:** 100% user adoption within 6 months of deployment
- **Onboarding Completion Rate:** Achieve 95% successful onboarding completion rate
- **Trial Conversion Rate:** Achieve 70% conversion rate for trial customers
- **Customer Time to Value:** Reduce time from order to service activation by 40%
- **Manual Application Processing Time:** Process 95% of manual FNO applications within 4 hours
- **Escalation Resolution Rate:** Resolve 90% of escalated tasks within defined SLA timeframes
- **FNO Reference Tracking Accuracy:** Achieve 100% accuracy in capturing and tracking FNO reference numbers

10. Risk Assessment

10.1. Technical Risks

- **FNO Integration Complexity:** Different FNOs may have varying API capabilities and requirements
- **Data Migration:** Migrating existing order data from legacy systems
- **Performance:** Ensuring system performance under high load

10.2. Business Risks

- **User Resistance:** Potential resistance to change from existing users
- **Training Requirements:** Extensive training may be required for complex features
- **Regulatory Compliance:** Ensuring compliance with telecommunications regulations

10.3. Mitigation Strategies

- Conduct thorough analysis of FNO requirements during design phase
- Implement comprehensive testing strategy including load testing
- Develop detailed training programs and change management plan
- Engage legal and compliance teams early in the development process

11. Conclusion

The ISP Order Management System represents a significant opportunity to modernize and streamline order processing operations. By implementing this comprehensive solution, the organization will be able to improve operational efficiency, enhance customer satisfaction, and position itself for future growth. The system's modular architecture and integration capabilities will provide the flexibility needed to adapt to changing business requirements and technological advances.

The success of this project will depend on strong collaboration between business stakeholders, technical teams, and external partners. Regular communication, thorough testing, and a phased implementation approach will be critical to ensuring a successful deployment and adoption of the new system.

The inclusion of a comprehensive Customer Onboarding System as a core component of the OMS ensures that new customers receive proper guidance and support during their initial service experience. The system's trial customer management capabilities will be critical for converting prospects into paying customers, with automated reminders and targeted conversion campaigns designed to maximize the conversion rate before the 30-day trial period expires. This integrated approach to order management and customer

onboarding will create a seamless experience from initial order creation through successful service activation and customer retention.